

Setting up NPVE GIS control for the ORION Nanofab

This procedure outline how to configure the NPVE software for automatic single GIS or Right Nozzle control and, if present, fully manual left nozzle control.

This document refers only to NPVE v4.6.2.722 and Fibics Remote Server v5.1.5.149 configured with ZEN Cairo v 7.9.0.16333. Older versions are not currently supported.

Setting up GIS Positions

Establish Nozzle Position list in ZEN:

1. Top 5 positions on ZEN list can be used by NPVE. Retracted and Standby should be created anywhere thereafter.
 2. Number Nozzle positions (1-5) then restart Athena server for ZEN to re-order list.
- Generate Nozzle position list in NPVE to match updated ZEN nozzle position list:

3. NPVE > GIS tab > Edit GIS States > Edit Nozzle Positions

Name	Position	
Retracted	0	Primary
Standby	0	Primary
W stage tilted @ 54 deg	1	Primary
W no tilt	2	Primary
C stage tilted @54 deg	3	Primary
C no tilt	4	Primary
PMCPs no tilt	5	Primary
XeF2	-1	Secondary

Figure 1: NPVE GIS Nozzle Position Editor

4. Duplicate position list that exists in ZEN. For each nozzle position available is ZEN:
 - i. Using ZEN, move nozzle to selected position
 - ii. In NPVE, click Add to add the nozzle position if necessary. Change its Name to match the name in ZEN
 - iii. Click the Read Current Position button and enter the reported position value into the appropriate cell in the Position column.
 - iv. Select the GIS controller associated with this nozzle position. For single GIS system, select Primary. The secondary controller cannot be controlled automatically
5. If a second GIS nozzle is present, it must be controlled manually though the Omni software. Any GIS state or Nozzle position associated with the Secondary nozzle will result in manual control mode when used in NPVE.
6. Click OK

Setting up GIS States

Configure the NPVE GIS states to reflect target nozzle positions and gas channels.

7. NPVE > GIS tab > Edit GIS States

State Name	Channel	Nozzle Position	Pos
W tilted w GaFIB	3	W stage tilted @ 54 deg	Primary
C tilted w GaFIB	1	C stage tilted @54 deg	Primary
PMCPS w GFIS (0 tilt)	2	Unspecified	Primary
XeF2 (left-manual)	-1	XeF2	Secondary

Figure 2: NPVE GIS States Editor

8. Add a desired GIS state. Each state will be used by NPVE Operations to control both nozzle position and gas channel.
9. Create GIS states in the GIS States Editor. For each GIS State:
 - i. Use ZEN to activate desired gas channel
 - ii. In NPVE, Click the Read Current Channel button and enter the corresponding number in the Channel column
 - iii. Select the nozzle position you want associated with this GIS state. The Pos column will indicate whether that nozzle position is in the controlled (Primary) or manual (Secondary) assembly.
10. Click OK.
11. Associate GIS States to specific FIB or GFIS operations through their respective Operation Editors. GIS States associated with the Secondary position will prompt the user for all GIS related intervention. States on the Primary position can be controlled automatically through Zen.

Additional Notes

- NPVE GIS setting rely on ZEN and must be reconfigured each time a change is made to the ZEN GIS position list. The Athena server must be restarted each time is made to the ZEN position list prior to reconfiguring the NPVE GIS settings.